

ThreatInsight: Innovating Early Threat Detection Through Threat-Intelligence-Driven Analysis and Attribution

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Abstract—The complexity and ongoing evolution of Advanced Persistent Threats (APTs) compromise the efficacy of conventional cybersecurity measures. Firewalls, intrusion detection systems, and antivirus software, which are dependent on static rules and predefined signatures, are increasingly ineffective against these sophisticated threats. Moreover, the use of system audit logs for threat hunting involves a retrospective review of cybersecurity incidents to reconstruct attack paths for attribution, which affects the timeliness and effectiveness of threat detection and response. Even when the attacker is identified, this method does not prevent cyber attacks. To address these challenges, we introduce ThreatInsight, a novel early-stage threat detection solution that minimizes reliance on system audit logs. ThreatInsight detects potential threats by analyzing IPs captured from HoneyPoints. These IPs are processed through threat data mining and threat feature modeling. By employing fact-based and semantic reasoning techniques based on the APT Threat Intelligence Knowledge Graph (APT-TI-KG), ThreatInsight identifies and attributes attackers. The system generates analysis reports detailing the threat knowledge concerning IPs and attributed attackers, equipping analysts with actionable insights and defense strategies. The system architecture includes modules for HoneyPoint IP extraction, Threat Intelligence (TI) data analysis, attacker attribution, and analysis report generation. ThreatInsight facilitates real-time analysis and the identification of potential threats at early stages, thereby enhancing the early detection capabilities of cybersecurity defense systems and improving overall threat detection and proactive defense effectiveness.

Index Terms—Advanced persistent threats, attacker attribution, HoneyPoint, knowledge graph, threat intelligence, threat knowledge.

I. INTRODUCTION

AS GLOBAL interconnectivity deepens and information technology rapidly advances, APTs have become a prominent cybersecurity concern. Orchestrated primarily by professional hacker groups or state-backed entities, APTs target critical

infrastructure, government sectors, and businesses to steal sensitive information or gain political leverage [1]. Characterized by their stealth, complexity, and durability, these threats involve sophisticated, multi-phased strategies including extensive intelligence gathering, tailored malware, and exploitation of zero-day vulnerabilities, enabling prolonged clandestine operations and continuous penetration of targets [2].

Detection techniques reliant on static signatures and rules, like firewalls and antivirus programs, are insufficient for countering evolving cyber threats. The complexity of APTs demands adaptive, innovative security measures. Provenance graphs, derived from system audit logs, are increasingly pivotal for nuanced threat detection. These logs record kernel activities, offering critical insights into attack dynamics within compromised systems. Provenance graphs elucidate the interactions and causal relationships among system elements such as processes, files, and sockets, illuminating attack pathways and mechanisms [3]. In dynamic attack-defense scenarios, attackers exploit system vulnerabilities, while defenders refine their strategies by studying adversarial tactics. Continuous monitoring and regular updates to security practices are essential [4]. Nonetheless, defenders face challenges in acquiring comprehensive threat knowledge, leading to gaps caused by limited understanding of adversaries. Bridging these gaps necessitates leveraging existing cyber threat knowledge. TI involves the gathering, analysis, and dissemination of information to identify and anticipate threats, providing crucial insights that bolster defensive measures [5]. Within the TI framework, Indicators of Compromise (IoCs) [6] are vital, representing observable suspicious activities that suggest potential threats to hosts or networks. The Pain Pyramid model [7] ranks these indicators by their acquisition difficulty and temporal stability, with common types including file hashes (e.g., files tied to specific malware) and IP addresses (e.g., IPs associated with particular attacks or command and control centers). Early identification of these IoCs enables organizations to adopt more proactive defense strategies, thereby enhancing their capabilities in threat mitigation.

Motivation: Threat hunting based on system audit logs models various types of IoC data to understand and identify attack behaviors. Recent advancements have incorporated TI to enhance log-based threat hunting. For instance, [8] introduced an automated framework integrating open-source TI, augmenting threat behavior pattern matching in system logs through TI-constructed

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threat behavior graphs. [9] and [10] link specific attack behaviors with broader semantic representations of APT Tactics, Techniques, and Procedures (TTPs) [11], generating detection rules for each attack phase. [12] synthesizes TTPs from various attack stages to create detailed attack pattern graphs for specific APT scenarios. Despite these advancements, current research primarily focuses on post-event analysis, challenging early threat detection and prevention. The effectiveness of these methods largely depends on audit logs containing comprehensive IoC data. While some studies have incorporated TI to expand log-based threat hunting, these methods remain predominantly reactive, mainly used for post-event retrospection rather than leveraging TI for proactive early detection and continuous monitoring. This reactive approach limits the effectiveness of defense response in the system's defense lifecycle. APT attack lifecycles typically involve multiple stages over extended periods, yet current detection methods often assume rapid completion of each phase to compromise the target system. [13] By the time attacks are detected, significant damage have usually been done.

Moreover, current research primarily targets the identification of attack behaviors from IoC data, often neglecting the crucial task of attributing attackers' identities from these data points. According to the QiAnXin TI Knowledge Pyramid model [14], identifying the attacker's identity represents both the most valuable and the most challenging type of knowledge to acquire within TI. Accurate attribution is vital for enacting proactive defense measures. Precise identification of the attacker allows for the comprehensive extraction of threat knowledge related to the attacker, enabling a detailed analysis of their motives, objectives, behavioral patterns, and technical methods. This in-depth analysis is crucial for revealing the strategic intentions of attackers and for effectively implementing deception, blocking, and additional countermeasures.

Challenge: In this work, we aim to achieve the following objectives: 1) effectively capture early-stage IoCs and extract relevant threat knowledge from TI; 2) design algorithms that model the threat knowledge associated with early-stage IoCs, accurately identifying and attributing attackers; 3) automatically aggregate early-stage IoCs and their associated threat knowledge, providing insights for subsequent defense strategies. Addressing these objectives presents three major challenges. First, acquiring early-stage IoCs is challenging, as they often include IPs and hashes prevalent in both legitimate and suspicious network activities, making it difficult to distinguish them and potentially leading to false positives and unnecessary blocks. Second, the effective mining and modeling of threat knowledge, as well as accurate attribution, are also challenging. Although early-stage IoCs, represented by IPs and hashes, are widely available and easy to collect, their raw form offers limited value. The real challenge lies in extracting meaningful threat knowledge and effectively modeling these characteristics to develop robust learning models. The model should not only accurately identify threats but also address the complexities introduced by data class imbalances, accurately recognizing minority class attacker samples to avoid overfitting and reduce false positives. Finally, our goal is to develop a real-time automated system capable of continuously capturing early-stage IoCs. This

system will leverage prior threat knowledge of TI to conduct in-depth analyses of early threats. By accurately uncovering potential attacker identities and comprehensively characterizing their threat profiles, we aim to ensure the timely provision of knowledge-driven, actionable threat defense insights.

In this study, we introduce the ThreatInsight system to tackle early threat detection challenges. Our key contributions are as follows:

a) Solution for Early Threat Detection: ThreatInsight introduces a novel detection scheme tailored for network environments. Utilizing decoy mechanisms called "HoneyPoint", it simulates legitimate services on external networks to capture the IP addresses of covert attackers during initial access. These IPs are analyzed using the TI Data Analysis Platform, which integrates external TI databases and a Neo4j database of APT-TI-KG, for expert-based associative threat knowledge mining. This process involves querying threat data related to IPs and their associated file hashes, thereby integrating threat knowledge of early IoCs. These features train an attacker attribution model based on APT-TI-KG semantics to predict attackers' identities. The predicted results automatically generate analysis reports for operational personnel, enhancing early threat detection and providing threat profiles to support defense strategy deployment and threat tracking.

b) Algorithm for Attacker Attribution: We developed an adaptive soft-voting ensemble learning model that integrates Random Forests (RF) [15] and Gradient Boosting Decision Trees (GBDT) [16]. This approach outperforms neural network baselines in overfitting resistance and minority class recognition. The model's accuracy and robustness are further enhanced through hierarchical weighted threat representations and SMOTE [17] data augmentation.

c) Dynamic Threat Perception Strategy: We propose a strategy considering attackers' complexity, resource levels, and TI data timeliness. This strategy provides insights into evolving network threats and a method for quantifying threat data value.

d) Effective and Practical System: Comprehensive evaluations using real attack data from cybersecurity defense practices show ThreatInsight's efficacy in attacker attribution and real-time operation. ThreatInsight accurately identifies attacker identities from threat feature data (macro-F1: 0.8343, micro-F1: 0.8561), significantly outperforming our best neural network baseline model (macro-F1: 0.2464, micro-F1: 0.4788). Additionally, the hierarchical weighted enhancement and SMOTE data augmentation strategies improved the $RG_{ensemble}$ model's macro-F1 and micro-F1 scores by 0.0512 and 0.0158, respectively. In terms of runtime performance, ThreatInsight processes approximately 400 distinct IPs daily. Dataset construction and model training, took total of 22.963 seconds and 1443.427 seconds respectively. The real-time analysis stage involved regex parsing of IPs, taking 1.162 seconds. TI data analysis using VT had an average response time of 2.667 seconds per IP query and 1.602 seconds per IP-associated file query. Model inference averaged 0.031 seconds per sample, and report generation took 0.689 seconds per report. For the worst-case runtime, assuming each IP is associated with 5 files (a rare case), querying 400 IPs in VT totals 4270.8 seconds. Adding model inference and

report generation times, the total worst-case time for processing these IPs is approximately 4559.962 seconds (about 76 minutes), ensuring efficiency even under challenging conditions.

The remainder of this paper is structured as follows: Section II reviews related work concerning KGs, attacker attribution techniques, and the utilization of the VT database. Section III describes the system architecture of ThreatInsight, detailing its design principles and the methodologies employed for each module. Section IV presents a comprehensive experimental evaluation of the ThreatInsight system. Finally, Section V concludes the paper and summarizes our findings.

II. RELATED WORK

ThreatInsight integrates APT-TI-KG with the VT database to enable comprehensive extraction of TI data linked to early-stage IoCs. APT-TI-KG serve as the essential data and semantic foundation for threat knowledge discovery, facilitating the modeling of threat features to infer attacker identities. Core analytical processes in ThreatInsight include integrated TI data extraction, semantic modeling, and threat knowledge discovery. This section accordingly reviews related works on Knowledge Graph, VT database, and Cyber Threat Attribution.

A. Knowledge Graph

KGs originated in the 1960s with the development of expert systems and semantic networks [18], [19]. KGs have since evolved significantly, utilizing the Resource Description Framework (RDF) [20] for structured knowledge representation. The advent of the World Wide Web and the Semantic Web further expanded KGs' applications [21]. Google's introduction of KGs in 2012 [22] exemplified their potential for intelligent services. Today, KGs are integral to AI, advancing semantic networks, Natural Language Processing (NLP) [23], and database technologies [24].

KGs are essential in intelligent services, as demonstrated by systems like YAGO [25], DBpedia [26], Freebase [27], and Probase [28], which enhance search engines, Question-Answering (Q-A) systems, and semantic analysis. In cybersecurity, KGs map complex relationships and reveal deep threat semantics [29]. They analyze security entities, identify threat patterns, and trace attack chains, thereby enhancing cybersecurity awareness and response capabilities. For instance, CSKG4APT integrated scattered intelligence and adjusts defense strategies proactively [30]. Ji et al. [31] introduced a knowledge-based Q-A system to enhance threat knowledge acquisition. Gao et al. [32] proposed a TI modeling system based on heterogeneous information networks, improving threat detection. Banerjee et al. [33] investigated data poisoning attacks on KGs, assessing their impact on KG integrity and potential defenses. In cybersecurity education, interactive platforms used KGs to teach attack and defense techniques [34]. Security operations centers utilized KGs for efficient threat detection [35]. The release of CDTier dataset has advanced Chinese TI Analysis [36]. In industrial control system security, KGs underpin data-driven defense frameworks, addressing complex threats by integrating multi-source threat data with network layouts [37].

In summary, by linking data from diverse semantic dimensions, KGs facilitate the management and utilization of knowledge. Integrating KGs with cybersecurity enhances the ability to analyze and infer implicit threat knowledge from vast datasets, supporting the development of more effective threat detection and defense strategies.

B. VT Database

VT database [38], a renowned online analysis platform, integrates over 70 antivirus tools to provide comprehensive TI services. Capable of analyzing URLs, IPs, and binary files [39], it is crucial for security assessment and threat hunting in complex network environments.

VT database's real-time detection and vast data resources are vital for cybersecurity research, especially in the identification of malicious entities. Kevin van Liebergen et al. [40] used VT's dataset to enhance large-scale threat hunting through clustering, while Zhu et al. [41] explored label aggregation for malicious tags, emphasizing threshold selection. Wang et al. [42] analyzed VT's dynamic labeling, offering insights and improvement suggestions for cybersecurity. Numerous studies have utilized VT to enhance threat detection accuracy and efficiency. Trivedi et al. [43] examined millions of spam samples in VT database, uncovering a significant presence of malware, particularly Trojans and ransomware, in email attachments. Roy et al. [44] explored the detection of elusive malicious URLs on Twitter, emphasizing platform vulnerabilities and the need for precise detection methods. Euijin Choo et al. [39] analyzed billions of URL scan reports, proposing a machine learning method to improve the accuracy of malicious URL identification and classification.

In summary, the effective utilization and analysis of VT's rich data resources have led to the development of innovative detection methods and tools. These studies have enhanced the precision and efficiency of detecting malicious entities and contributed to a deeper understanding and prevention of cyber threats.

C. Cyber Threat Attribution

Cyber threat attribution transforms data into actionable knowledge by extracting relevant associations from vast and dispersed threat data, linking attack behaviors to responsible entities. This process begins with the efficient collection and processing of multi-source, heterogeneous data using data engineering techniques. Intelligent pattern mining and correlation analysis then reveal critical threat knowledge, uncovering the origins and motivations behind attacks.

Data engineering forms the foundation of cyber threat attribution, especially in the collection, processing, and management of multi-source heterogeneous data. Given the complexity of network environments and the diversity of attack techniques, effective threat attribution depends on various data sources, such as network traffic data, system logs, and TI. Bilge et al. [45] highlighted the critical role of preprocessing network traffic data, especially in identifying botnet command-and-control servers via NetFlow analysis. Techniques like data cleaning, feature

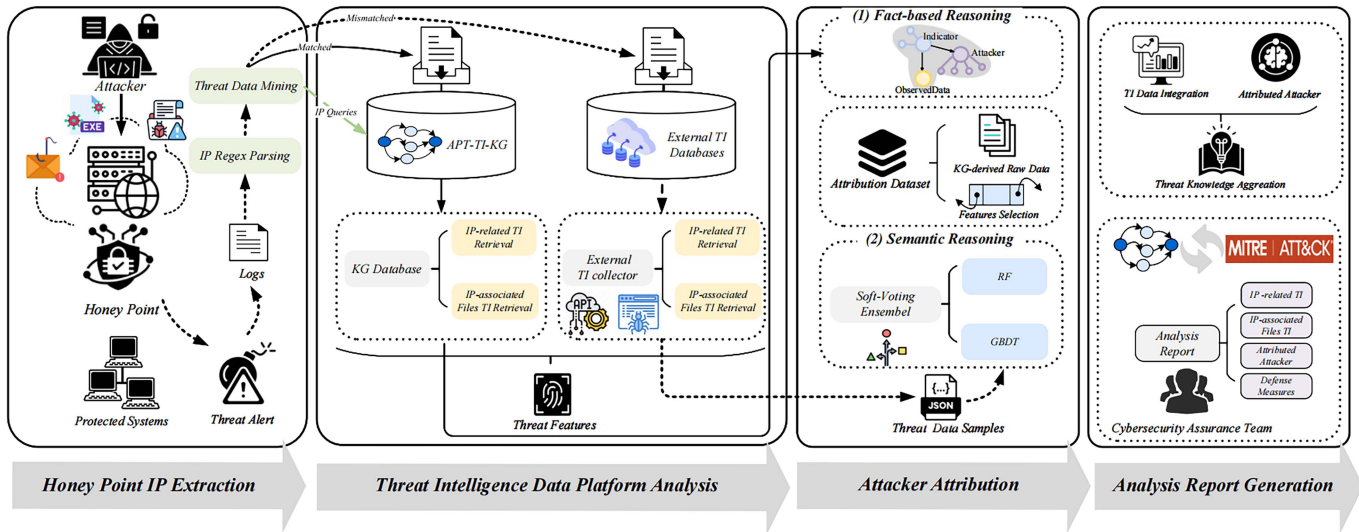


Fig. 1. Overview of the ThreatInsight system architecture.

extraction, and normalization are essential to ensure the accuracy and robustness of attribution analysis. Additionally, the collection and analysis of TI are crucial to cyber threat attribution. Integrating multi-source TI data allows for comprehensive attacker profiling and pattern recognition. Thonnard [46] used clustering techniques on large-scale intelligence data to identify attack trends, while Irshad et al. [47] applied NLP and machine learning techniques to attribute threat entities from unstructured intelligence reports. These studies underscore the importance of effective TI data collection and analysis in adapting to evolving threats and refining attribution methods.

Behavior pattern analysis is crucial in cyber threat attribution by extracting and analyzing key threat features from complex datasets and transform them into actionable knowledge. Following feature extraction, pattern recognition techniques classify threat entities by analyzing similarities between different threat features. Benthin [48] demonstrated automated attribution by analyzing malware binary features and tracking variants using machine learning algorithms, laying the groundwork for identifying consistent attack patterns. Alrabaee et al. [49] employed binary code similarity analysis to trace the origins of malwares. Bayer [50] and Perdisci [51] used behavioral clustering to group malwares with similar behaviors. Chai et al. [52] addressed the limitations of traditional clustering methods with the Dynamic Prototype Network (DPNSA), which dynamically adapts to unknown malware by updating behavior prototypes. The application of pattern analysis extends beyond malware classification. Gao et al. [53] introduced a real-time method for querying streaming system monitoring data to detect anomalies in enterprise systems, improving real-time monitoring and anomaly detection accuracy. Similarly, Mazzawi et al. [54] proposed a behavioral pattern analysis approach to identify malicious activities in large databases, demonstrating the broader applicability of behavior pattern analysis in diverse cybersecurity contexts.

In summary, cyber threat attribution is fundamentally a deep threat knowledge discovery process. It focuses on identifying

key threat information through data-driven behavior correlation and pattern analysis, building a comprehensive understanding of threat actors, and ultimately enhancing defense and response capabilities in a knowledge-driven manner.

III. THREATINSIGHT SYSTEM ARCHITECTURE

System Overview: The ThreatInsight system is an innovative framework for early-stage threat analysis and attribution, designed to enhance detection capabilities and operational efficiency. The system architecture, as illustrated in Fig. 1, consists of four key modules: 1) *HoneyPoint IP Extraction*, which identifies IPs interacting with HoneyPoint, serving as the initial data point for subsequent analysis; 2) *TI Data Platform Analysis*, which integrates APT-TI-KG with external TI databases to conduct comprehensive queries on the identified IPs and their associated files, providing a holistic investigation of IP-related activities; 3) *Attacker Attribution*, which employs an ensemble learning algorithm to analyze the TI data and infer attacker identities; and 4) *Analysis Report Generation*, which compiles the analysis results into comprehensive reports with threat analysis and recommended defense measures.

In essence, cyber threat attribution involves linking cyberattacks to potential attackers based on the traces left by the attackers. ThreatInsight has the ability to deduce probable assailants and produce analytical reports by utilizing minimal "traces" during the initial phases of a cyber threat. In practical terms, within ThreatInsight, HoneyPoint IP Extraction is responsible for obtaining the IPs associated with early threats - the "traces" left by covert attackers through HoneyPoint. TI Data Platform Analysis manages the retrieval of TI data pertaining to these IPs, uncovering the "clues" associated with these "traces". Attacker attribution involves training models based on APT-TI-KG data, then using these "clues" as inputs to predict the identity of the attacker through the trained model. Finally, the Analysis Report Generation module integrates the findings from the previous

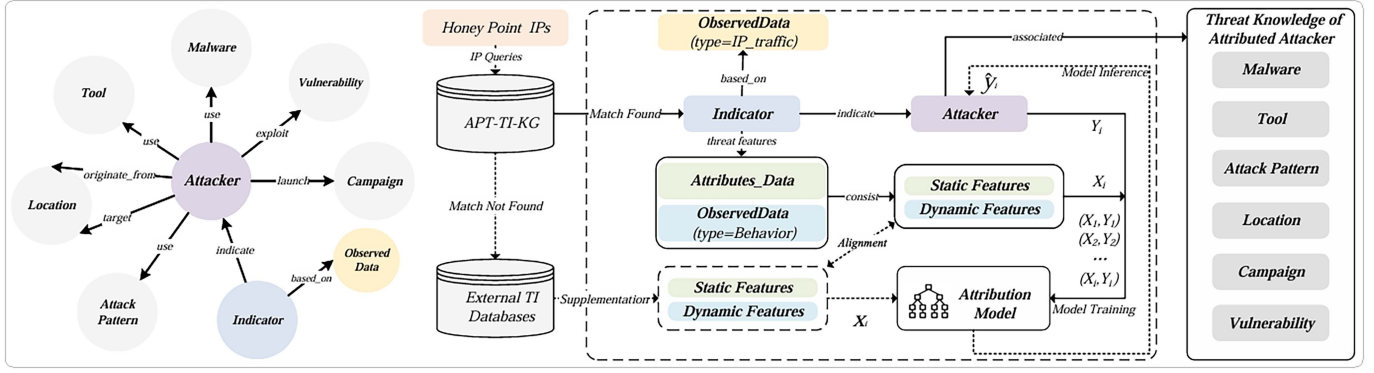


Fig. 3. APT-TI-KG ontological architecture and its utilization in ThreatInsight. Outlines the structural design and integration of the APT-TI-KG within the ThreatInsight system, highlighting its role in TI analysis and attacker attribution.

For each successfully matched $ip_m \in IP_{matched}$, retrieve the associated TI data from the *ObservedData* nodes' attributes and query the related *Indicator* nodes to obtain TI data on malicious files:

$$OD_{ip} = \{od \in OD_{kg} \mid od.ip = ip_m\} \quad (3)$$

$$IND_{file} = \{ind \in IND_{kg} \mid ind \rightarrow od, od \in OD_{ip}\} \quad (4)$$

The TI data for IPs and associated malicious files' static features is extracted as follows:

$$TI_{ip} = \{data \mid data \in Attributes(OD_{ip})\} \quad (5)$$

For each $ind_f \in IND_{File}$, retrieve its attributes and related behavioral *ObservedData* for file TI data:

$$OD_{beh} = \{od \mid od.type = beh, ind_f \rightarrow od\} \quad (6)$$

$$TI_{file} = \{data \mid data \in Attributes(IND_{file}, OD_{beh})\} \quad (7)$$

The complete TI data is aggregated as:

$$TI_{KG_analysis} = \bigcup_{IP_m \in IP_{matched}} (TI_{ip} \cup TI_{file}) \quad (8)$$

This method systematically extracts and analyzes TI data from captured IPs, offering comprehensive KG-based TI data analysis.

2) *Analysis of VT Database*: The KG analysis method enables the retrieval of IP-related TI data but has limitations: the *Indicator* nodes within the KG represent only a limited subset of IoCs. For IPs that do not find matches within the KG, ThreatInsight employs an external TI database to supplement the TI data.

Specifically, define the set of unmatched IPs $IP_{unmatched}$ as follows:

$$IP_{unmatched} = IP_{hp} \cap IP_{matched}^c \quad (9)$$

where IP_{hp} denotes the set of IPs captured by the HoneyPoint, and $IP_{matched}$ refers to the set of IPs successfully matched in the KG analysis.

The VT database analysis process is formalized as a series of structured query tasks, denoted by $Q_{VT}(IP_u)$ for each $IP_u \in$

$IP_{unmatched}$. Each task $Q_{VT}(IP_u)$ comprises the following steps:

- 1) Obtain the IP report $R_{IP}(IP_u)$ for IP_u .
- 2) Extract TI data $TI_{ip}(IP_u)$ from $R_{ip}(IP_u)$.
- 3) Retrieve the set of malicious file hashes associated with the IP: $H_{mal}(IP_u)$.
- 4) For each $h \in H_{mal}(IP_u)$, obtain its file report $R_{file}(h)$.
- 5) From $R_{file}(h)$, extract static attributes $TI_{stat}(h)$ and dynamic behavioral information $TI_{behav}(h)$.
- 6) Integrate $TI_{ip}(IP_u)$ with $TI_{stat}(h)$ and $TI_{behav}(h)$ for all related files to compile the comprehensive TI data $TI_{VT}(IP_u)$.

Following this sequence of analysis queries, we obtain the supplementary VT TI data for each IP in $IP_{unmatched}$.

$$TI_{VT} = \bigcup_{IP_u \in IP_{unmatched}} TI_{VT}(IP_u) \quad (10)$$

By combining the TI data obtained from the KG analysis and VT database analysis, we construct a comprehensive TI data framework for the original input IP_{hp} :

$$TI_{total} = TI_{KG_analysis} \cup TI_{VT} \quad (11)$$

where $TI_{KG_analysis}$ originates from the previous analysis process on the KG. TI_{total} will serve as the data foundation for subsequent automated report generation and attack trace analysis.

In the TI data analysis platform, the retrieved TI data is aggregated by IP as the primary key and stored in JSON format for subsequent module retrieval and utilization. The retrieved malware TI data is categorized into two types: static attribute information and dynamic behavior information. Static attribute information includes file size, creation date, hashes, etc., while dynamic behavior information is derived from analyzing the sequence of behaviors exhibited by the malware. The detailed TI data of IP and IP-associated files are presented in Tables III, IV, and V in Appendix A, available online.

3) *Complexity Analysis of Queries: TI Analysis of APT-TI-KG*: This process begins by matching IPs with *ObservedData* nodes within the KG, entailing a complexity of $O(n \times m)$, where n represents the number of IPs and m denotes the *ObservedData* nodes. This step incurs high computational

costs, particularly with large node counts. For each matched IP, TI data is retrieved from an average of p *Indicator*, leading to a complexity of $O(k \times p)$, with k being the successfully matched IPs. To streamline real-time batch queries, we have implemented a hash table that directly maps IPs to KG nodes, reducing the lookup complexity to $O(n)$ with constant lookup time $O(1)$ per IP. This enhancement not only boosts query efficiency but also mitigates the performance impact of node scalability. The integration of hash tables simplifies the matching process, rendering the system adept at managing large-scale, real-time data flows while preserving analytical precision. The space complexity for storing IP-related TI data is $O(n + k \times p)$, where n is the number of IPs, k is the number of matched IPs, and p is the average number of *Indicator* nodes per IP. The use of hash tables for IP-to-KG node mappings also requires additional space of $O(n)$. *TI Analysis of VT Database*: This method involves external API queries for unmatched IPs, with complexity of $O(u)$, where u is the count of unmatched IPs. The extraction and integration of malicious file hashes and other TI data for each IP yield a complexity of $O(u \times q)$, with q representing the associated files per IP. Despite potential delays from external API calls, the overall process complexity is relatively modest, dependent mainly on the unmatched IP count and file associations. The space complexity is $O(u + u \times q)$, where u is the number of unmatched IPs and q is the number of associated files per IP.

4) *TI Collector for Multi-Source External TI Integration*: ThreatInsight employs a TI collector to integrate data from diverse external sources, extending IP analysis capabilities beyond traditional platforms like VT to include AnHeng Starmap [60] and QiAnXin TI platform [61]. This method enhances the comprehensiveness of IP examinations by utilizing a wider range of TI sources.

Utilizing API programming and web scraping, the collector retrieves IPs, associated malicious file hashes, and their respective TI from multiple databases, effectively addressing potential data gaps in single-source systems. Additionally, analysts can create customized templates to align the analysis of these databases with specific cybersecurity scenarios, ensuring data consistency and fulfilling precise analytical needs.

C. Attacker Attribution

1) *Fact-Based Reasoning*: The ThreatInsight system employs the APT-TI-KG for attacker attribution by analyzing matched IPs through previously retrieved *Indicator* nodes during the TI Data Platform Analysis. Specifically, once the *Indicator* nodes associated with matched IPs are identified, the system extends its queries to the *Attacker* nodes indicated by these *Indicator* nodes. This targeted querying efficiently completes the attacker attribution process by leveraging factual relationships within the APT-TI-KG.

Complexity Analysis: The complexity of this fact-based attacker attribution process primarily depends on the number of *Indicator* nodes associated with each matched IP, denoted by i .

Given that each IP's *Indicator* nodes have already been identified in a previous analysis phase, the complexity of querying the *Attacker* nodes is directly proportional to the number of these indicators. This results in an overall time complexity of $O(i)$, which is more efficient compared to multi-layered querying across different node types in the KG. The space complexity for storing the *Indicator* and *Attacker* nodes data is $O(i + a)$, where a represents the number of *Attacker* nodes. This space complexity is manageable and ensures efficient storage and retrieval during the attribution process.

2) *Semantic Reasoning*: In the APT-TI-KG, for IPs not directly linked with factual relationships, we model the TI data obtained from the VT database to predict attacker identities. This process leverages the semantic information from the *Indicator* nodes previously retrieved during the TI Data Platform Analysis phase.

The *Indicator* nodes encapsulate static features of the malicious files associated with IPs, while interactions between these nodes and adjacent *ObservedData_{behavior}* nodes reveal dynamic behavioral characteristics, providing information about the behaviors and quantifying their frequency through edges. This combination of static and dynamic threat features assists in inferring attacker identities through machine learning techniques.

Dataset Construction for Attacker Attribution: Attacker attribution is framed as a multi-classification challenge aimed at predicting attacker identities indicated by *Indicator* nodes based on their static and dynamic features. As depicted in Fig. 3, the static and dynamic features of *Indicator* nodes are represented as X_i , while the labels indicating attacker identities are denoted as Y_i . Consequently, the data samples are structured as pairs (X_i, Y_i) , with the training objective being the prediction of Y_i . The KG is conceptualized as a directed graph $G = (V, E)$, where V represents the set of nodes, and E the set of edges connecting these nodes. The primary focus lies on *Indicator* nodes V_{ind} and their relationships $E_{i,o}$ with *ObservedData* nodes. For each *Indicator* node n in V , the dataset construction proceeds as follows:

- 1) Extract attribute data $Attr_{ind}$ from node n .
- 2) For behavior-type *ObservedData*, collect behavioral data and its occurrence frequency B_{info} from the edges connecting n to n .
- 3) Identify the potential attackers $A_{attacker}$ indicated by n .
- 4) Consolidate $Attr_{ind}$, B_{info} , and $A_{attacker}$ into the dataset.

This methodology ensures the creation of a comprehensive dataset combining both static and dynamic features, facilitating the training of machine learning models to accurately predict potential attacker identities.

Complexity Analysis of Dataset Construction: The complexity of the dataset construction algorithm for the ThreatInsight system is chiefly determined by the number of *Indicator* nodes in the APT-TI-KG. For each indicator node, the algorithm conducts several operations including attribute extraction, accessing connected *ObservedData* nodes of behavior types, and identifying associated attackers. The complexity of these operations is largely linear relative to the number of

Indicator nodes, as the number of linked *ObservedData* and *Attacker* nodes per indicator is generally small and constrained. Consequently, the overall time complexity is $O(|V_{\text{ind}}|)$, where $|V_{\text{ind}}|$ denotes the total number of indicator nodes processed. The space complexity for storing the extracted attributes, connected *ObservedData* nodes, and related attacker information amounts to $O(|V_{\text{ind}}| + |V_{\text{obs}}| + |V_{\text{att}}|)$, where $|V_{\text{obs}}|$ represents the number of *ObservedData* nodes and $|V_{\text{att}}|$ the number of *Attacker* nodes. This linear complexity is deemed low, facilitating efficient processing even as node counts increase. Importantly, this dataset construction occurs during the pre-deployment phase, completing during the system setup and not affecting the real-time performance of the operational system. This preparation ensures that the system remains responsive and efficient during active threat monitoring, free from the burdens of ongoing data processing.

Attacker Attribution Algorithm: This study extends foundational machine learning methods for classifying malicious activities by employing the TI data of the APT-TI-KG for multi-class attacker attribution. To tackle the inherent challenges of multi-class complexity and data imbalance, where *Attacker* nodes are unevenly linked to *Indicator* nodes, we implement the Synthetic Minority Over-sampling Technique (SMOTE). This technique is particularly tailored to enhance the representation of under-represented classes. We adjust them to the median class count, based on their initial distribution. Specifically, SMOTE is configured with $k = 5$ nearest neighbors to generate synthetic samples, ensuring a balanced distribution across classes. In the preprocessing phase, non-indicative features such as hash values are excluded due to their collision-resistant properties, which render them non-informative for deriving preimages. Based on expert experience, differential weights refine threat characterization: *mutex_operation* is weighted the highest at 3, while *registry_key_operation*, *service_operation*, and *files_operation* receive a weight of 2. Lesser critical behaviors are assigned a weight of 1, enhancing the significance of threat characterization. The enriched dataset, comprised of both static and dynamic features of malicious files, is used to train a model that efficiently predicts attacker identities, forming a robust mechanism for attacker attribution that is resilient to the skew inherent in the original practical data.

Integrated Ensemble Model for Attacker Attribution: This study employs a sophisticated ensemble model combining RF and GBDT to enhance the accuracy and generalizability of attacker attribution. RF operates through an ensemble of decision trees, each trained on a random subset of the dataset, which introduces diversity and robustness in the model [15]. This method mitigates imbalances in datasets by reducing variance through bootstrap sampling and random feature selection at each node split. The RF model is constructed by averaging the predictions from B trees for each sample x , calculated as: $RF(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$. GBDT, on the other hand, sequentially corrects errors from previous iterations by fitting new trees to the residuals of the loss function [16]. Starting from an initial model F_0 , GBDT refines the prediction across M steps, progressively enhancing the model by adding a fraction

of the new tree's prediction, $\nu h_m(x)$, to the existing model $F_m(x) = F_{m-1}(x) + \nu h_m(x)$.

The combination of RF and GBDT into a single ensemble model leverages the strengths of both techniques: RF's robust handling of diverse data features and GBDT's focused error correction. The predictions from both models are integrated using a soft voting mechanism, where the outputs are weighted based on their confidence levels:

$$Y_{\text{ensemble}}(x) = \alpha \cdot RF(x) + (1 - \alpha) \cdot GBDT(x). \quad (12)$$

Here, α is adjusted to optimize the balance between RF and GBDT, enhancing the ensemble's ability to handle various aspects of the data effectively. This soft voter approach is preferred over hard voting as it allows the ensemble to consider the probabilistic confidence of each model's prediction, rather than just their categorical outcomes, providing a more nuanced and effective decision-making process. This method not only increases the precision of the predictions but also ensures that the ensemble adapts dynamically to the strengths and weaknesses of each model across different scenarios, thereby increasing the reliability and accuracy of attacker attribution.

3) *Dynamic Threat Perception Strategy:* Malicious activities, such as scanning or brute-force attacks, often display periodic or irregular patterns that evolve due to changes in attackers' infrastructure and the nature of malicious files over time. These changes are influenced by the attackers' resource levels and capabilities. To address the varying relevance of historical TI associated with an IP, a dynamic threat perception strategy is implemented. This strategy modulates inference results by integrating temporal data and the evolving capabilities of attackers.

The core of this strategy is the Dynamic Threat Perception Function (DTPF), which diminishes the influence of older malicious file data in threat assessments, reflecting the decreased reliability of historical data over time:

$$Y_{\text{perception}}(\Delta t, \lambda) = e^{-\Delta t \cdot \lambda} \quad (13)$$

where Δt represents the time elapsed since the last data analysis, and λ is a decay factor that correlates with the attacker's resource intensity and threat level. The decay factor adjusts based on the attacker's resources and capabilities, which dictate the frequency of infrastructure and file updates. This factor is quantified using the *resource_level* and *sophistication* fields from the QiAnXin TI Platform and expert annotations, typically rated from 1 to 5. Higher values signify greater resources or capabilities, necessitating quicker updates to reflect current threat landscapes. The function employs the mean of these metrics to determine λ 's appropriate value, ensuring the strategy aligns with the dynamic nature of cybersecurity threats.

D. Analysis Report Generation

The Analysis Report Generation module automates the synthesis of analysis results into comprehensive, actionable reports for cybersecurity experts. It leverages data processed by preceding modules, organizing it into a Microsoft Word document using Python's *docx* library. This involves extracting data from

JSON files and formatting it for readability. The module aggregates critical information on IPs, associated malicious files, behavioral analysis outcomes, and potential APT attacker attributions. This aggregated data forms the core of the report, which comprises key sections: 1) IP-related TI consolidates TI about the IP; 2) Attributed Attackers details attackers linked to the IPs as identified by the attacker attribution module; 3) IP-associated Malicious Files TI covers both static and dynamic features of the malicious files connected to the IPs; 4) Recommended Defense Measures suggests defense strategies against the TTPs of the attributed attackers. Each report offers a comprehensive overview from initial data entry to detailed threat analysis, ensuring that the insights are both thorough and immediately useful for enhancing security measures.

IV. EXPERIMENTAL EVALUATION

We developed ThreatInsight using Python. Neo4j and VT were integrated for TI queries. Sklearn and imbalanced-learn were employed for the attacker attribution pipeline and SMOTE oversampling, with the python-docx library for report generation. Deployed on HoneyPoint servers, ThreatInsight collects covert attackers' IPs, facilitating threat awareness analysis and performance evaluation. Table VI in Appendix B, available online outlines the deployment environment. We conducted four experiments to evaluate ThreatInsight's efficacy and efficiency. The first assessed the attacker attribution algorithm's performance against baselines. The second measured the efficiency of each ThreatInsight module. The third tested the DTPF's effectiveness and parameter tuning. The fourth demonstrated ThreatInsight's analysis process using two IP instances captured in the HoneyPoint, and presents ThreatInsight's partial analysis results during major security assurance activities.

A. Attacker Attribution Evaluation

1) *Experimental Setup:* a) *Dataset:* The dataset for the attacker attribution model is constructed from APT-TI-KG, which integrates open-source TI data collected and annotated by our team with real attack sample data from cybersecurity defense practices. The dataset comprises a total of 823,716 nodes and 3,303,178 edges. We constructed the dataset based on Indicators of IPs and Files types, linked to their corresponding Attacker nodes. The dataset includes 4,258 samples across 33 attacker categories. Fig. 11 in Appendix B, available online shows the imbalanced distribution of data across these attacker categories.

b) *Evaluation Protocol:* We performed stratified sampling on the dataset, dividing it into training and validation sets in a 9:1 ratio. A 10-fold cross-validation was employed to evaluate the model's overall performance. The final reported performance metrics, including F1-score, Recall, and Precision, represent the model's overall performance across all folds. These metrics are averaged using both Micro-average and Macro-average methods to provide a comprehensive evaluation [62].

Let C be the total number of classes, and TP_i , FP_i , and FN_i [63] denote the true positives, false positives, and false negatives for class i , respectively.

The Micro-average method aggregates TP, FP, and FN across all classes to compute overall performance metrics, emphasizing effectiveness on majority classes. The specific calculations are:

$$Precision_{micro} = \frac{\sum_{i=1}^C TP_i}{\sum_{i=1}^C (TP_i + FP_i)} \quad (14)$$

$$Recall_{micro} = \frac{\sum_{i=1}^C TP_i}{\sum_{i=1}^C (TP_i + FN_i)} \quad (15)$$

$$F1_{micro} = \frac{2 \cdot Precision_{micro} \cdot Recall_{micro}}{Precision_{micro} + Recall_{micro}} \quad (16)$$

The Macro-average method calculates performance metrics for each class individually and then averages them, giving equal weight to each class and reflecting balanced performance. The specific calculations are:

$$Precision_{macro} = \frac{1}{C} \sum_{i=1}^C \frac{TP_i}{TP_i + FP_i} \quad (17)$$

$$Recall_{macro} = \frac{1}{C} \sum_{i=1}^C \frac{TP_i}{TP_i + FN_i} \quad (18)$$

$$F1_{macro} = \frac{1}{C} \sum_{i=1}^C \frac{2 \cdot Precision_i \cdot Recall_i}{Precision_i + Recall_i} \quad (19)$$

These evaluation methods provide a comprehensive understanding of the model's performance on imbalanced datasets, examining both its balanced performance across all classes and its overall effectiveness.

c) *Implementation of Models:* We compared ThreatInsight's $RG_{ensemble}$ of attacker attribution model with several baseline models based on deep neural networks and decision trees. The implementation details are as follows:

- *Long Short-Term Memory (LSTM)* [64]: LSTM is a type of Recurrent Neural Network (RNN) that captures long-term dependencies in sequential data. L2 regularization and Dropout layers were used to mitigate overfitting. The model consists of a fully connected layer, Batch Normalization layer, and an output layer with softmax activation. It was compiled using the Adam optimizer and sparse categorical cross-entropy loss.
- *Feedforward Neural Network (FNN)* [65]: FNN is a basic neural network where information flows in one direction. Our FNN includes two fully connected layers and a Batch Normalization layer, with Dropout layers to mitigate overfitting. The output layer uses softmax activation. The model was compiled using the Adam optimizer and sparse categorical cross-entropy loss.
- *Multilayer Perceptron (MLP)* [66]: MLP is an FNN with at least one hidden layer. Our MLP has three fully connected layers and a Batch Normalization layer, along with two Dropout layers to mitigate overfitting. The output layer uses softmax activation. The model was compiled using the Adam optimizer and sparse categorical cross-entropy loss.

TABLE I
PERFORMANCE COMPARISON OF RG_{ENSEMBLE} AND BASELINE MODELS

Method	Model	Macro-Avg			Micro-Avg		
		F1	Recall	Precision	F1	Recall	Precision
Neural Network	MLP	0.1177	0.1093	0.1274	0.3553	0.3553	0.3553
	FNN	0.1766	0.1715	0.1821	0.4368	0.4368	0.4368
	LSTM	0.2464	0.2419	0.2511	0.4788	0.4788	0.4788
Decision Tree	RF	0.6980	0.6561	0.7456	0.7954	0.7954	0.7954
	GBDT	0.7754	0.7369	0.8182	0.8314	0.8314	0.8314
Proposed	RG_{ensemble}	0.7831	0.7441	0.8265	0.8403	0.8403	0.8403
	$RG_{\text{ensemble_BW}}$	0.7911	0.7513	0.8353	0.8424	0.8424	0.8424
	$RG_{\text{ensemble_SMOTE}}$	0.8238	0.7967	0.8528	0.8530	0.8530	0.8530
	$RG_{\text{ensemble_BW_SMOTE}}$	0.8343	0.8059	0.8648	0.8561	0.8561	0.8561

- **RF:** Random Forest is an ensemble learning method that combines multiple decision trees for prediction. We constructed a forest with 450 trees, requiring at least 2 samples to split an internal node, 1 sample per leaf node, and no maximum tree depth.
- **GBDT:** Gradient Boosting is an ensemble method that iteratively adds models to correct the previous step's errors. Our GBDT model has a learning rate of 0.1, a maximum tree depth of 3, at least 1 sample per leaf node, 10 samples per internal node, and a total of 500 trees.
- **RG_{ensemble} :** Based on the configurations of the RF and GBDT models, ThreatInsight constructs an adaptive soft-voting classifier, RG_{ensemble} , with a parameter α set to 0.8. RG_{ensemble} serves as the basic model proposed in this study. $RG_{\text{ensemble_BW_SMOTE}}$ is the model trained on the dataset enhanced with Behavior-Weighted (BW) features. $RG_{\text{ensemble_BW_SMOTE}}$ is the model trained on the dataset augmented using the SMOTE oversampling technique. $RG_{\text{ensemble_BW_SMOTE}}$ is the model trained on a dataset incorporating both behavior-weighted features and SMOTE augmentation.

2) **Evaluation Results:** As shown in Table I, the following observations can be made: 1) **Lower Performance of Neural Network Models:** Neural network models exhibit lower performance on this dataset, likely due to their complexity leading to overfitting. Despite using Dropout, regularization, and batch normalization, the generalization ability on the test set remains insufficient. For instance, the best-performing neural network model, LSTM, shows training and validation losses as depicted in Fig. 13 in Appendix B, available online. Training loss stabilizes after an initial drop, while validation loss increases, indicating overfitting. Additionally, LSTM's Micro metrics are significantly higher than its Macro metrics, suggesting the model is biased towards learning features of majority classes and ignoring minority classes, highlighting sensitivity to data imbalance. Similar tendencies are observed in other neural network baselines. 2) **Better Performance of Decision Tree Models:** Compared to neural networks, RF and GBDT perform better on this dataset. 3) **Performance Improvement of RG_{ensemble} :** Our RG_{ensemble} method, utilizing a soft voting ensemble strategy, shows significant performance improvement over baseline models. Specifically, RG_{ensemble} outperforms the LSTM model by 0.5367 and 0.3615 in Macro-F1 and Micro-F1 scores, respectively; RF by 0.0851 and 0.0449; and GBDT by 0.0077

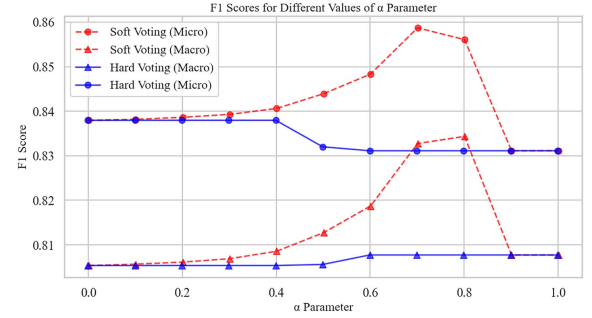


Fig. 4. Micro-F1 and Macro-F1 of $RG_{\text{ensemble_BW_SMOTE}}$ with soft and hard voting under different α parameter settings.

and 0.0089. Additionally, RG_{ensemble} with feature weighting ($RG_{\text{ensemble_BW}}$) and SMOTE oversampling ($RG_{\text{ensemble_SMOTE}}$) further enhance performance. Ultimately, $RG_{\text{ensemble_BW_SMOTE}}$ improves Macro-F1 and Micro-F1 scores by 0.0512 and 0.0158 than RG_{ensemble} , respectively. 4) **Mitigation of Class Imbalance:** Both decision tree models and $RG_{\text{ensemble_SMOTE}}$ show more significant improvements in Macro-F1 compared to Micro-F1, indicating effective model optimization and data augmentation in enhancing learning and prediction of minority classes, thus alleviating class imbalance to some extent.

Furthermore, we evaluated the performance of the soft and hard voters of $RG_{\text{ensemble_BW_SMOTE}}$ with different α parameter settings (using 0.1 increments). As shown in Fig. 4, the experimental results indicate that although the scores of soft voting and hard voting are similar or identical for certain α values, soft voting generally outperforms hard voting in both Macro-F1 and Micro-F1 scores. Specifically, soft voting achieves the highest F1 scores at α values of 0.7 and 0.8 for Micro and Macro metrics, respectively. These findings suggest that by considering the predicted probabilities of classifiers, soft voting can more effectively enhance overall classification performance. We selected the α parameter with the highest Macro-F1 score (0.8) for reporting the performance of the soft voter $RG_{\text{ensemble_BW_SMOTE}}$.

B. ThreatInsight System Efficiency Evaluation

During our team's participation in critical cybersecurity assurance activities, we typically process approximately 400 distinct IPs daily. Consequently, we evaluated the operational efficiency of ThreatInsight's daily task load, which includes early threat

TABLE II
THREATINSIGHT SYSTEM EFFICIENCY EVALUATION

Task	Operation	Response Time (s)	Availability
HoneyPoint IP Extraction	Regex Parsing	1.162	N/A
APT-TI-KG Query	Hash Table Lookup (IP)	1.227×10^{-6}	1.000
	KG Lookup ($ObservedData_{ip}$)	0.914	1.000
	KG Lookup ($ObservedData_{behavior}$)	0.914	1.000
	KG Lookup (<i>Indicator</i>)	0.914	1.000
VT Database Query	IP Query	2.667	0.985
	IP-associated File Query	1.602	0.973
Attacker Attribution	Dataset Construction	22.963	N/A
	Model Training	1443.427	N/A
	Model Inference	0.031	N/A
Report Generation	Analysis Result Integration	0.689	N/A

analysis and attribution for these 400 IPs. As shown in Table II, we present the efficiency evaluation results for each module based on the various stages of ThreatInsight deployment and application: 1) *Preparation*. This stage involves constructing the attacker attribution dataset and model training. The dataset construction includes retrieving attacker nodes and associated *Indicator* nodes, *ObservedData* nodes, and edge information from the APT-TI-KG storage using Cypher queries, which takes a total of 22.963 seconds. Following dataset construction, model training under 10-fold cross-validation takes approximately 1443.427 seconds (24 minutes) in total. 2) *Real-Time Analysis*. a) *HoneyPoint Log Parsing*: This phase involves the regular expression parsing of IPs from log files, taking a total of 1.162 seconds. b) *TI Data Analysis*: After obtaining the IPs, they are input into the TI data platform for TI correlation queries. For VT database, the average response time for each IP query is 2.667 seconds, and for each IP-associated file query, it is 1.602 seconds. The efficiency of external TI database analysis is influenced by API rate limits and real-time network speed, making query data susceptible to network fluctuations. Availability is calculated by the ratio of successful responses to total queries. In the analysis of 400 IPs using VT, the availability of IP TI queries is 0.985, and the availability of IP-associated file TI queries is 0.973. The availability metrics indicate that a small fraction of queries did not receive timely responses, which could affect the comprehensiveness of TI data collected in real-time scenarios. To enhance ThreatInsight's robustness and reliability, we implemented a checkpoint mechanism to ensure timely storage of intermediate data. This approach helps to minimize service disruption during interruptions and allows efficient recovery and continuation of processes from the last saved state. The efficiency of TI data analysis based on APT-TI-KG is relatively high. The time to query specific IP values for $ObservedData_{ip}$ nodes is 0.914 seconds. This process is optimized by constructing a hash table from the $IP_traffic$ type $ObservedData_{ip}$ in the KG, significantly reducing the average query time per IP to 1.228×10^{-6} seconds. After matching the IP with $ObservedData_{ip}$ nodes, further queries for associated *Indicator* nodes take an average of 0.914 seconds. Further querying on the behavior type of the related *Indicator* node in $ObservedData_{behavior}$ takes an average of 0.914 seconds. 3) *Model Inference*. The data obtained from the TI data platform

is subsequently input into the attacker attribution model for prediction. The average prediction time for each data sample is 0.031 seconds. 4) *Report Generation*. This stage also involves aggregating the analysis data produced by the upstream modules and querying the APT-TI-KG for TI related to the top-3 attackers identified by the attribution model. The final step is compiling these results into a report file, with the average time to generate each analysis report being 0.689 seconds.

To assess the worst-case run-time performance of the ThreatInsight system, we conducted the following estimates based on processing 400 IPs under the most challenging conditions, where all queries are directed to the external VT database, as it has significantly higher latency compared to the APT-TI-KG analysis. And all IPs undergo semantic reasoning. For *real-time analysis*, regex parsing of IPs from log files takes 1.162 seconds. For *TI data analysis*, the average response time for each IP query in the VT database is 2.667 seconds, and for each IP-associated file query, it is 1.602 seconds. Assuming each IP is associated with 5 files (a number observed in only a few cases during practice), the total time for querying 400 IPs is $400 \times (2.667 + 5 \times 1.602) = 4270.8$ seconds. The *model inference* involves predicting the data samples obtained from the TI data platform, with an average prediction time of 0.031 seconds per sample, resulting in a total of 12.4 seconds for 400 IPs. For *report generation*, aggregating the analysis data and generating each analysis report takes an average of 0.689 seconds, totaling 275.6 seconds for 400 reports. Summing these times, the *total worst-case time* for processing 400 IPs is approximately 4559.962 seconds (about 76 minutes).

C. Effectiveness Analysis of DTPF

We validated the effectiveness of the proposed perception function outlined in Section III-C3. In the conducted experiments, as shown in (2), Δt represents the time interval between the current and the most recent analysis of data samples, indicating the recency of TI. The decay factor λ , derived from the *resource_level* and *sophistication* fields in the KG, determines the rate of weight decay, reflecting the attacker's threat level and resource strength.

Fig. 5 illustrates the impact of different Δt and λ values on the perception function. The simulations covered Δt values from 0

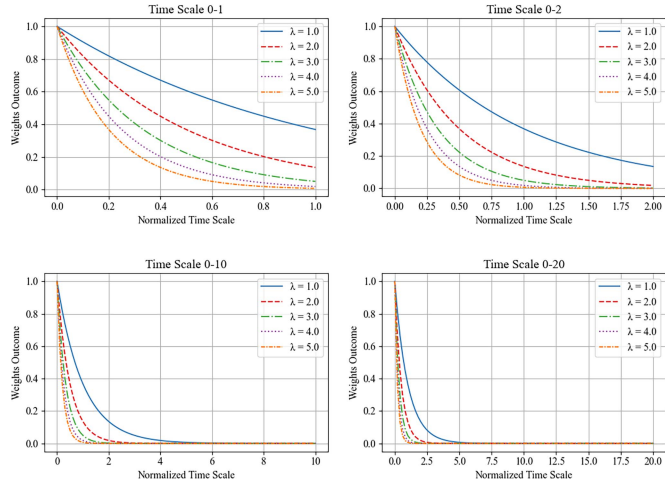


Fig. 5. Simulated DTPF weight outcomes across various time span Δt and decay factor λ .

to 20 and λ values ranging between 1 and 5. Results confirm accelerated weight decay for higher λ values, aligning with the concept that attackers with greater resources and sophistication frequently update their infrastructure and malicious files. Additionally, the function demonstrates a progressive decrease in weight with longer time spans, showcasing its dynamic ability to adjust weights based on the perception of timeliness in TI. Notably, weight changes tend to stabilize beyond certain time intervals, particularly evident in the 0-10 and 0-20 simulations. This underscores the importance of selecting an appropriate time span interval for practical applications. Analysis suggests that a 0-2 range is optimal for normalizing timestamp differences, with the 0-1 range particularly effective for capturing weight variations without the diminishing influence of λ observed over longer intervals.

D. Case Study

1) *Comparison of Analysis : APT-TI-KG Versus VT Database:* The IP analysis methodology, utilizing both APT-TI-KG and the VT database, presents a detailed case study of two specific IPs. This case study highlights the differences in data analysis and processing workflows between both methodologies.

APT-TI-KG Analysis: In this example, the IP 45.76.128.165 is analyzed, as shown in Fig. 6. In the KG, it corresponds to an *ObservedData* node of the *ip_traffic* type. The 1st Cypher query is constructed to match *Indicator* nodes based on the *ObservedData* node, providing detailed threat insights. Counting the behavioral edges from the *Indicator* node to related *ObservedData* nodes provides detailed data on behavior frequency, helping to shed light on the attacker's tactics. Fig. 12 in Appendix B, available online provides detailed descriptions of the nodes' attributes.

The 2nd Cypher query identifies APT33 as the attacker associated with the *Indicator* node, providing insights into the entity behind the IP. Further inquiries into APT33 uncover additional TI data and inform defense strategies based on the group's TTPs.

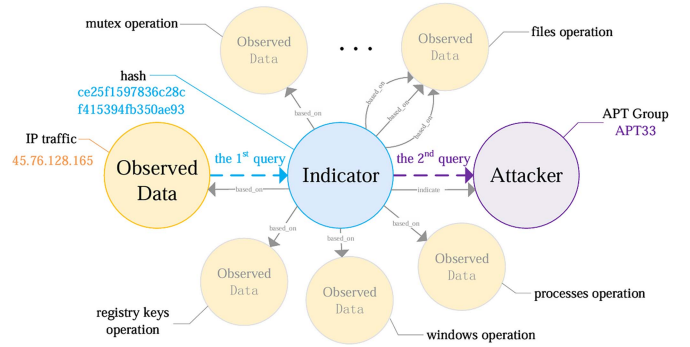


Fig. 6. Analysis of 45.76.128.165 Using the APT-TI-KG.

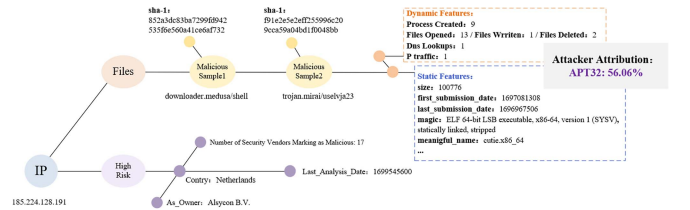


Fig. 7. Analysis of 185.224.128.191 using VT database.

Detailed attributes of the APT33 node are provided in Fig. 14 Appendix B, available online.

VT database Analysis: When IPs, like 185.224.128.191, are not found in the KG, the VT database is used for TI data queries. As shown in Fig. 7, VT analysis reveals that 185.224.128.191 is geolocated in the Netherlands and associated with the autonomous system *Alsicon B.V.* This IP is flagged as malicious by 17 security vendors.

Furthermore, VT analysis of malicious files associated with 185.224.128.191 reveals characteristic data. Fig. 7 shows feature information for two such files (Sample 1 and Sample 2). These analyses extract file TI from both static and dynamic features, consistent with the KG's analytical dimensions. By organizing these features, a data sample is constructed for the attribution model. This sample is processed by the attribution inference module, predicting APT32 as the likely attacker with a 56.06% confidence level, as shown in Fig. 8. The automated report generation module then aggregates the TI data and inference results, completing the threat profile for APT32. Detailed attributes of APT32 node are provided in Fig. 15 in Appendix B, available online.

2) *Cyber Threat Analysis of Major Events:* During the 19th Asian Games in Hangzhou and the 134th China Import and Export Fair, the ThreatInsight system provided real-time cybersecurity assurance. It collected IPs accessed from HoneyPoint and applied early threat detection algorithms for comprehensive assessment. IPs were categorized into two groups: those marked as malicious by security vendors and those without such labels. Immediate countermeasures were conducted for the former, while the latter underwent in-depth TI and attacker attribution analyses, followed by continuous monitoring by security operations teams. Throughout both events, the ThreatInsight system

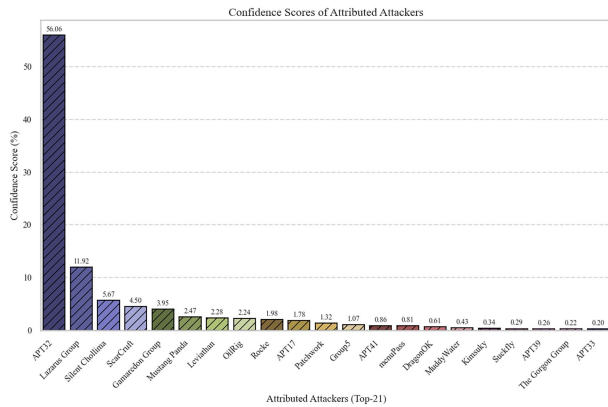


Fig. 8. Confidence (Top-21) of attributed APT groups for 185.224.128.191.

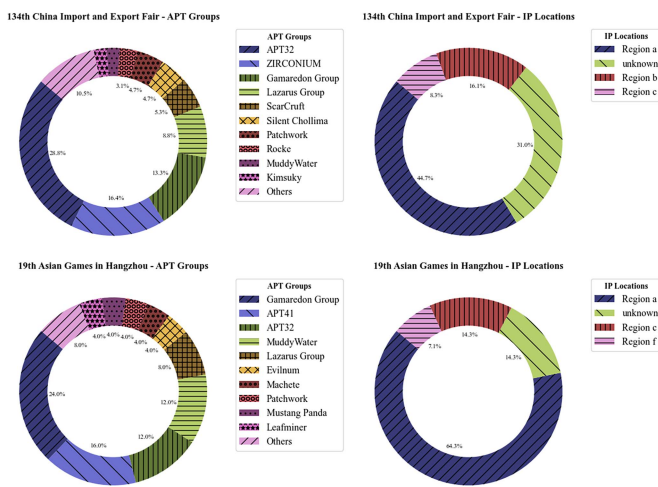


Fig. 9. Analysis of attacker attribution and geographical distribution of IPs during the 19th asian games in Hangzhou and the 134th China import and export fair.

evaluated over 4,300 IPs, generating real-time detailed reports and promptly disseminating findings to assurance teams for informed decision-making.

During these events, ThreatInsight's analysis, as shown in Fig. 9, uncovered potential APT activities involving groups such as APT32, ZIRCONIUM, and the Gamaredon Group. Notably, the geographical distribution of these threat IPs primarily originated from specific countries. By integrating ThreatInsight's insights into the operational scope and tactics of these APT organizations with real-time threat monitoring, analysts could infer the attackers' technical strategies, facilitating prompt and effective countermeasures. The implementation of ThreatInsight has greatly enhanced the analytical capabilities of cybersecurity teams. Its invaluable data support has played a pivotal role in strengthening comprehensive and timely cyber defenses in collaboration with supporting security departments.

V. CONCLUSION

The ThreatInsight system enhances cybersecurity defenses by innovatively detecting APTs through mining and modeling

early-stage IoC's related threat knowledge. This proactive approach expands the scope of threat detection, improves system responsiveness to emerging threats, and promotes a shift towards proactive defense strategies. As cyber threats evolve, the advancements enabled by ThreatInsight highlight the need for ongoing refinement in threat intelligence discovery and utilization. Our future research will focus on innovative approaches to utilizing threat knowledge for APT detection, aiming to refine threat hunting techniques and formulate more comprehensive defensive strategies.

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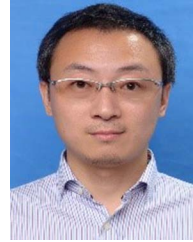


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